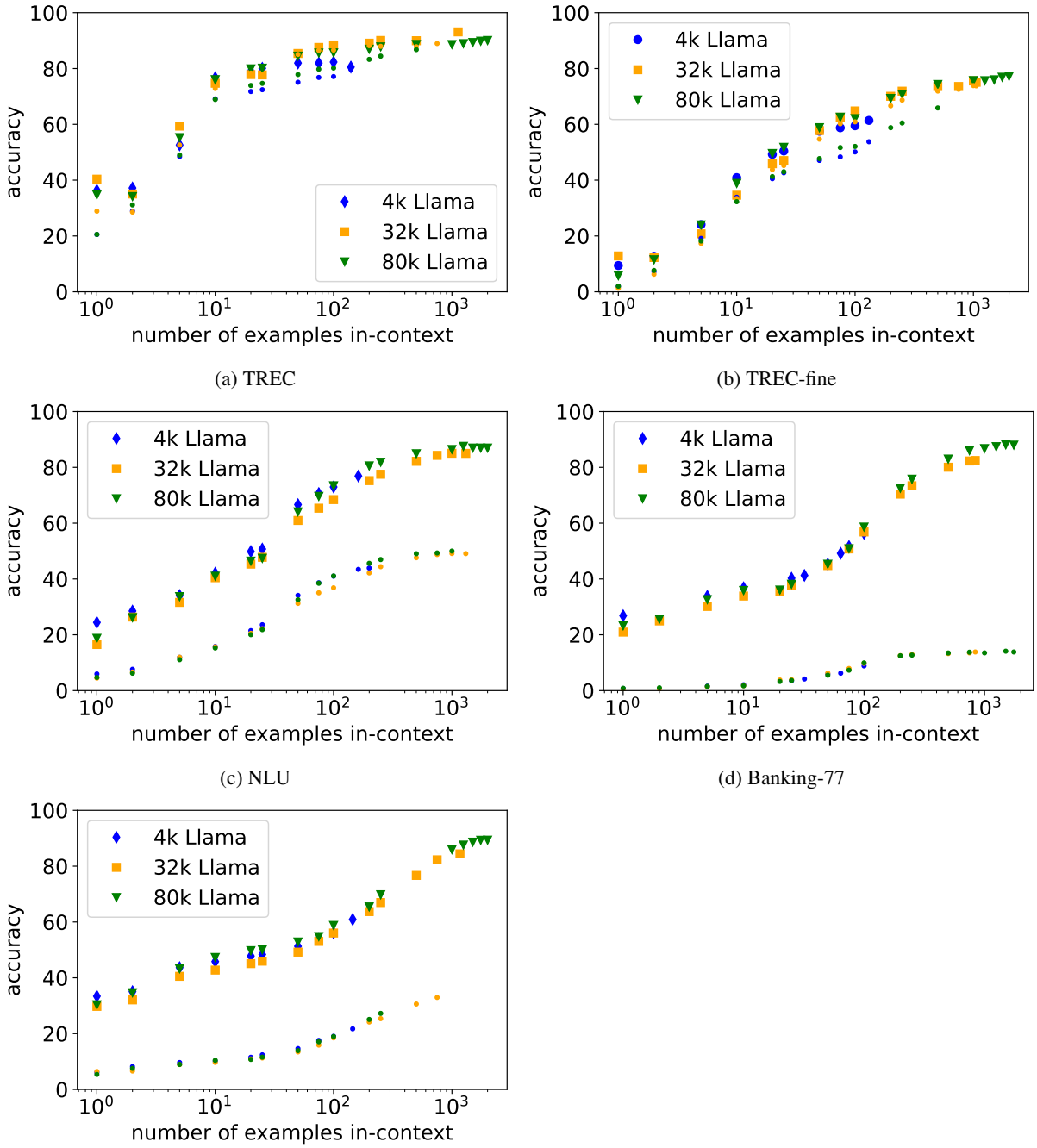
## D Constrained decoding details

To perform constrained decoding, following Ratner et al. (2022), we add a large constant to all logits at each decoding step that could result in a valid label being generated. This strategy is generally not possible for API-access models, or when the label space is not fully known in advance. To determine whether the same trends hold without the use of constrained decoding, we evaluated the Llama-2 family models with and without constrained decoding. Figure 14 shows the comparison; while performance is lower, especially in the higher-label-space datasets, the general trends hold. In Banking77, it appears that performance begins to saturate slightly earlier with unconstrained decoding; we hypothesize that this may be due to more specialized language used in banking domain labels.




(a) TREC

(b) TREC-fine

(c) NLU

(d) Banking-77

(e) Clinic-150

Figure 14: Comparison of constrained decoding (diamond, square, and triangle datapoints on the graph) with the unconstrained decoding variants (dot-like datapoints of the same colors).

## E Finetuning

We performed finetuning with HuggingFace transformers (Wolf et al., 2020). To perform parameter efficient finetuning (PEFT), we used the peft (Mangrulkar et al., 2022) package (version 0.9.0). We finetune the model for 30 epochs, evaluating it every epoch on the test set, and ultimately choosing the checkpoint with the highest test accuracy. We note that using the test set to perform model selection presents an unfair advantage to finetuning (compared to ICL) and may not be truly indicative of the generalization error. However, doing so provides the advantage of being both comparable to ICL in terms of the data used, as well as giving an upper bound on the true generalization accuracy of the finetuned model, further emphasizing any observed efficacy gap between it and ICL.

**Initialization of the classification head** While in our default setting we initialize the classification head from the pretrained LM head, subsampled at the representation of the first token in each label, we investigate the efficacy of this approach by contrasting with a randomly initialized classification head. Figure 15 shows that while in the few-shot regime, this approach has significant advantage, as the training set grows in size the difference shrinks to become negligible. In no case was random initialization better than this approach.

**Hyperparameter tuning** To remain comparable in terms of compute efficient finetuning, we did not perform extensive hyper-parameter tuning per task, and instead experimented with a good global setting on a single dataset (Banking-77). Specifically, we experimented with different learning rates, different LoRA ranks ($r$) and $\alpha$ (Hu et al., 2022) and also tried applying RSlora (Kalajdzievski, 2023) which sets the scaling factor to $\frac{\alpha}{\sqrt{r}}$ as some evidence suggest it can outperform the original method. Figure 16 summarizes the results, depicting average test accuracy against training examples with different settings.

Ultimately, we found that using HuggingFace's (Wolf et al., 2020) default parameters of $r = 8$, $\alpha = 32$, LoRA dropout of $0.1$ and a learning rate of $1e-3$ to work best. In all cases, we used batch sizes of 32 and weight decay of $0.01$.

It is possible that methods specialized for finetuning in small-data regimes, such as T-few (Liu et al., 2022), might close the gap between ICL and PEFT in the small-data regimes. We did not consider T-few in our analysis because of its additional pretraining stage, which imposes substantial additional cost, and because T-few was developed with a focus on encoder-decoder models and we consider only decoder-only models in our setting.

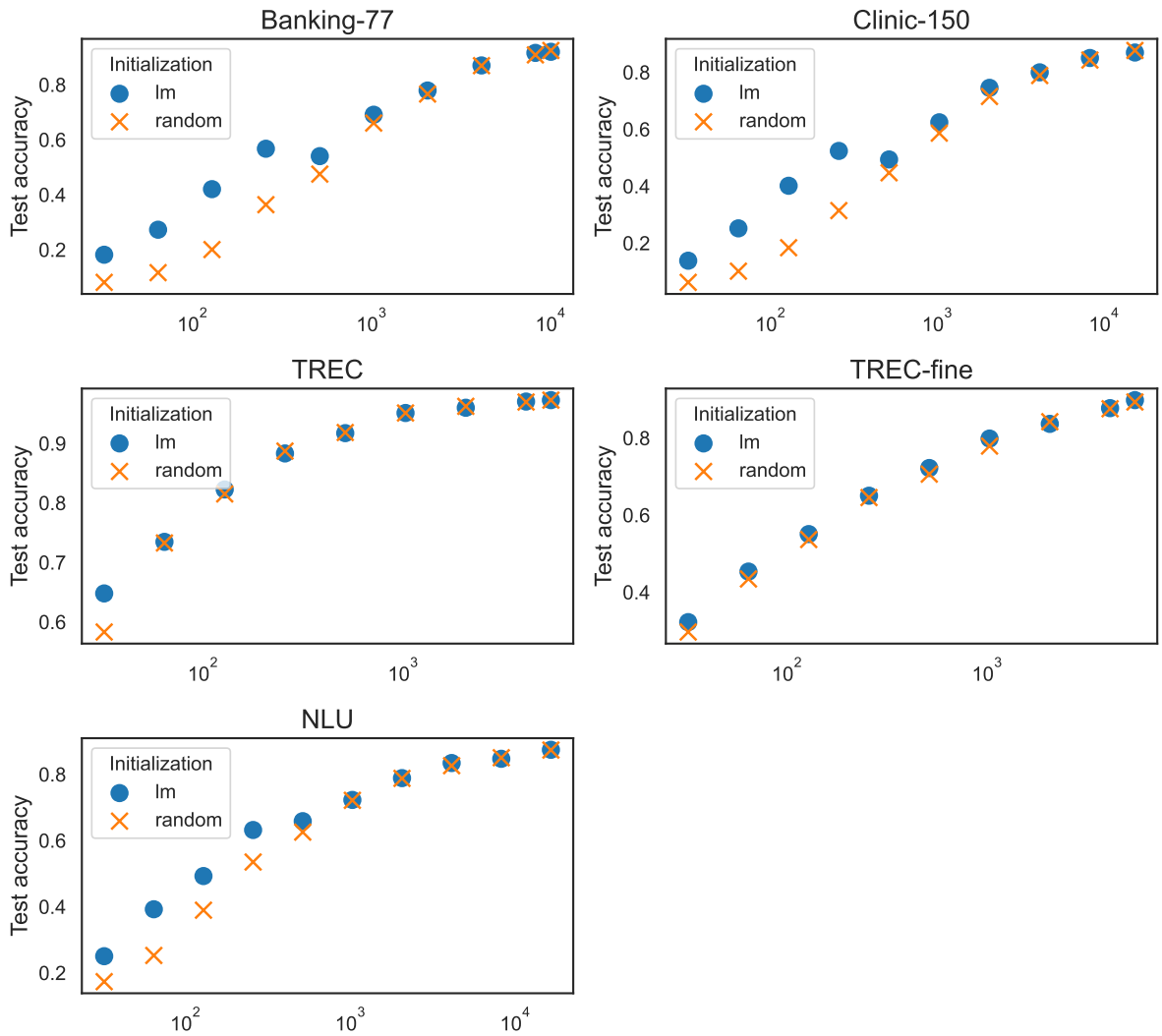



Figure 15: Comparing initialization methods of the classification head when finetuning a PEFT llama-2-7b model. Averaged (best) test accuracy over 5 random seeds. Initialization with *lm* subsamples the pretrained language-modeling head at the first token of the target label, while random samples random weights.

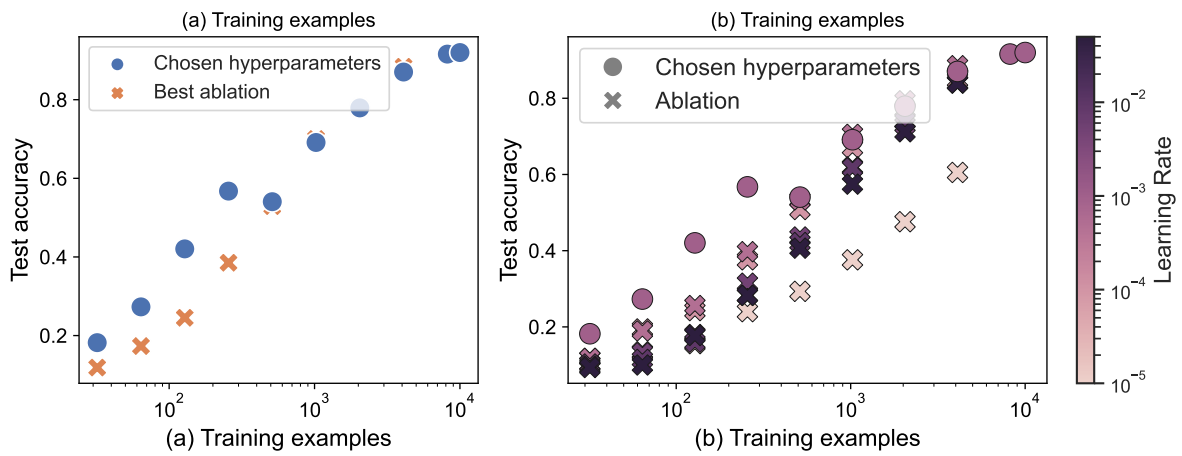



Figure 16: Comparing hyperparameters when finetuning a PEFT llama-2-7b model on Banking-77. Averaged (best) test accuracy over 3 random seeds. (a) Comparing our fixed LoRA configurations to the best alternative configuration (at each scale) we tried. (b) Comparing different learning rates.